Quantifying Medication “Off-Periods” in Parkinson’s Disease via Smartphone-Based Explainable Motor Symptom Analysis

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***Abstract*—Parkinson’s Disease (PD) is a progressive neurodegenerative disorder that affects more than 10 million people globally. One of the primary difficulties in dealing with it is the *wearing-off phenomenon* the time before the next Levodopa dose when the medication’s effectiveness diminishes and the motor symptoms return. Today, clinical monitoring relies mostly on patient diaries, which are not very effective in recording the detailed changes needed for accurate dose adjustments.**

We present PD-Monitora smartphone-only system that exploits three motor biomarkers validated clinically: resting tremor through a triaxial accelerometer, finger-tapping (bradykinesia index), and visual reaction time (pre-motor cognitive lag). This system performs assessment of a patient’s motor state pre and post medication. It further determines a *Medication Deviation Gap* ($\Delta_{\text{med}} = |S_{\text{pre}} - S_{\text{post}}|$) as a digital biomarker for off-period severity. A hybrid model combining Random Forest and Support Vector Machine (SVM), with Platt scaling incorporated for probability calibration of continuous severity scores, is presented in the case of *binary classification trap* that limits existing models to a Healthy/PD dichotomy. The pipeline is being trained on the mPower ResearchKit dataset, supplemented by 500 synthetic, iOS-specific samples generated using CTGAN (for tabular feature distributions) and TimeGAN (for temporal accelerometer sequences), and implemented via CoreML for privacy-preserving, on-device inference. The findings show the practicality of this setup as a personalized early-warning system that neurologists can use to help adjust patient treatments.

***Index Terms*—Parkinson’s Disease, Wearing-Off, Medication Off-Period, Digital Biomarker, Explainable AI, SVM, Random Forest, Smartphone, mPower, CoreML, Bradykinesia, Tremor, Welch’s PSD, React Native**

I. Introduction

Parkinson’s Disease is the second major neurodegenerative disorder in elderly, characterized by gradual destruction of dopaminergic neurons in the substantia nigra. The lack of dopamine leads to various motor symptoms, such as resting tremor (4–6 Hz) bradykinesia rigidity, and postural instability - symptoms together known as TRAP. Although drug treatment, especially the administration of Levodopa, can alleviate these symptoms, patients on long-term therapy usually develop a **wearing-off phenomenon**: in the progression of the disease, the levels of Levodopa in plasma drop 2 to 4 hours after each dose, and the motor symptoms come back even before the next dose is given.

The up and down swinging between the symptomatic*On-periods* and the distressing *Off-periods* greatly lower the quality of life of the patients and is the main reason for changing the Levodopa dose during the disease progression. At present, the disease monitoring mostly depends on the Movement Disorder Society Unified Parkinson’s Disease Rating Scale (MDS-UPDRS), which is performed by the trained neurologists during the rare outpatient visits, as well as on the patient-written symptom diaries in-between the periods. Both these methods have a common main limitation: they are sporadic, very subjective, and are not able to record the daily changes that are typical of the wearing-off experience.

A. The Binary Trap in Computational Research

Indeed, the computational literature is indicative of the same clinical gap. To a large extent, machine learning research on Parkinson's Disease (PD) has been focused on the simple task of binary classification—distinguishing a group of healthy individuals from a group of diagnosed PD patients. We call this the **Binary Trap**: a strict classifier is unable to depict the symptomatic continuum, misclassification is enforced for ambiguous cases and most importantly, it is of no practical use to an already diagnosed patient who needs ongoing medication effectiveness monitoring rather than a repetitive binary decision.

B. Research Contributions

The paper tackles the clinical and computational limitations by making the following contributions:

- An N-of-1 monitoring framework that keeps track of the patient's condition over time and shifts the focus from *diagnosis* to *measuring the effects of medication*;
- The Medication Deviation Gap (Δ_{med}) as a digital biomarker personalized and quantitative for the severity of wearing-off;
- A hybrid RF/SVM ensemble with Platt scaling that produces a continuous Severity Index and solves the Binary Trap; smartphone-only version with React Native and CoreML, which does not require costly wearable devices.

II. Literature Review

A. Digital Biomarkers for PD via Wearables and Smartphones

Wearable technologies have meaningfully advanced the field of PD monitoring. The WATCH-PD multi-center study validated Apple Watch sensors for detecting early PD progression, achieving sensitivity above 85% for tremor detection. Mahadevan et al. developed digital biomarkers for resting tremor and bradykinesia using a wrist-worn accelerometer, establishing the feasibility of consumer-grade IMUs for continuous PD monitoring. However, these systems have remained predominantly research-focused or clinician-oriented, and lack patient-centric interfaces suited to daily self-management. A further limitation is that most wearable tools generate outputs optimized for healthcare providers, leaving patients without the means to interpret their own symptom trends between clinical visits.

B. Smartphone-Based PD Studies and the mPower Dataset

The mPower study, conducted by Sage Bionetworks using Apple's ResearchKit framework, established one of the largest open-access mobile PD datasets, collecting accelerometer, tapping, voice, and walking data from 8,320 participants. Surangsrirat et al. applied the mPower tapping module to 1,851 participant samples and demonstrated that finger-tapping features could reliably separate participants into three statistically distinct PD severity clusters, each correlating significantly with MDS-UPDRS I–II and PDQ-8 scores. Their unsupervised K-means analysis identified *numberTaps*, *meanTapInter*, and *corXY* as the most discriminative features—findings that directly inform the tapping feature design in the present work.

Arroyo-Gallego et al. demonstrated that smartphone typing patterns could classify PD with 81% sensitivity and specificity. Prince and De Vos applied a deep learning framework to mPower sensor data, achieving strong binary classification performance. Schwab and Karlen combined walking, voice, tapping, and memory modules from mPower to construct a multi-modal classifier. While these studies collectively validate the platform, all remain confined to binary Healthy vs. PD discrimination.

C. Mobile Applications for PD: A Landscape Review

Estévez-Martín et al. conducted a systematic review of 92 PD applications across iOS and Android, finding that 34% were designed around periodic symptom monitoring. Tremor was addressed by 27% of reviewed applications, bradykinesia and slowed movement by 32%, and balance by 20%. Critically, no application was found that integrates pre- and post-medication motor assessment as a within-subject longitudinal biomarker—a finding that confirms the research gap this work is designed to address. Their review further established that finger-tapping tests are a well-validated and easily implementable proxy for motor speed, lateral coordination, and dopamine-sensitive bradykinesia.

D. LLM-Integrated PD Applications

Bhalala and Mansoor presented PARKA AI, an iOS application that integrates Apple Watch HealthKit data (tremor, gait, and sleep) with self-reported logs and Google Gemini 1.5 Flash to generate both patient-friendly summaries and structured HCP reports. While PARKA AI represents a significant step forward in patient-provider communication, it depends on Apple Watch hardware as an external sensor requirement and does not implement a quantitative medication deviation analysis. The present work is architecturally complementary: where PARKA AI provides LLM-mediated communication, PD-Monitor provides a physics-grounded, explainable ML biomarker for off-period detection using only the smartphone.

E. Research Gap

Taken together, the above body of work reveals a clear research gap: no prior smartphone-only system has been designed to quantify the *within-subject delta* between medicated and unmedicated motor states as a continuous, personalized biomarker for wearing-off. Existing studies are either cross-sectional (binary Healthy/PD), hardware-dependent (Apple Watch, APDM Opals), architecturally opaque (deep CNNs/LSTMs), or focused on general symptom logging rather than therapeutic monitoring. The proposed PD-Monitor directly fills this gap through a longitudinal N-of-1 design supported by an explainable ML pipeline.

III. System Architecture

PD-Monitor is a full-stack mobile health system organized into three tiers: a React Native mobile frontend, a Node.js/TypeScript backend, and a MongoDB Atlas persistence layer. ML inference is performed on-device via CoreML.

A. Mobile Frontend (React Native / Expo / iOS CoreMotion)

The client application is built with React Native and Expo, targeting iOS 14 and above. Three assessment tests are put into practice:

1) Rest Tremor Test: During the task, the patient either holds the phone motionless or paws the device on a flat surface for 30 seconds. iOS CoreMotion streams triaxial accelerometer data at 100 Hz, i. e. twice the 60 Hz rate imposed by browser-based sensor APIs are bypassed. Raw (t, x, y, z) tuples are kept in a `Float64Array` and sent to the backend for further processing.

2) Finger-Tapping Test (Bradykinesia Index): In this test, the patient is asked to alternately tap a large on-screen target with two fingers within 20 seconds. The mPower study protocol is followed in the process. With every tap, a timestamp of millisecond precision is recorded. The timelines of inter-tap intervals, tap-missing events, as well as freeze gaps (pauses > 500 ms) are calculated client-side before being sent over to server. Visual Reaction Time Test

3) Visual Reaction Time Test: A colored stimulus appears after a random delay of 1–3 seconds so as to prevent anticipation of the response. The time span between the appearance of the stimulus and the patient's tap is taken down for each round. Taking an average of five different trials results in a more precise estimation of the pre-motor latency.

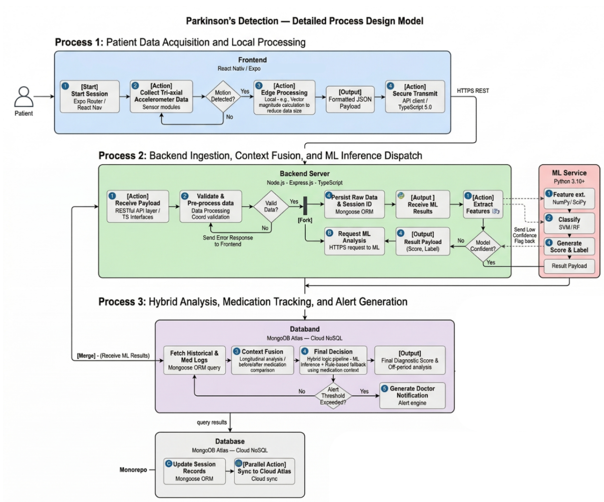

Fig. 1. PD-Monitor System Architecture: Three-tier design showing React Native frontend with CoreML inference, Node.js/TypeScript backend, MongoDB Atlas database, and secure data flow between components.

B. Backend (Node. js / Express. js / TypeScript)

Backend development revolves around Node. js, Express. js, and TypeScript. To prevent silent data corruptions in ML features, static typing is used to maintain schema validity of incoming medical data. Below is an example of a TypeScript interface:

```
interface TremorPayload {
  patientId: string;
  testPhase: "pre_medication"|
             "post_medication";
  timestamp: number;
  accelerometerData: {
    x: number[]; y: number[];
    z: number[];
    samplingRateHz: number;
  };
}
```

REST endpoints are in charge of managing test ingestion, fetching patient histories, and scoring deviations in off-periods. Feature extraction is conducted in Python and triggered on the server as a child process.

C. Database (MongoDB Atlas)

We picked MongoDB Atlas (NoSQL) because the variable-length JSON accelerometer arrays are not very compatible with relational row structures. Each patient document carries a rolling 30-day window of pre- and post-medication sessions, thus making it possible to calculate the baseline per-patient very efficiently without the use of costly JOIN operations.

D. On-Device ML Inference (CoreML)

The model is prepared offline in Python using scikit-learn, then it is changed to Apple's CoreML format through `coremltools`. Such deployment method comes with three major advantages: (1) **privacy**—the raw sensor data is never sent off the device during inference; (2) **low latency**—no network round trip is needed; and (3) **offline capability**—the assessments are still operational without an internet connection.

IV. Methodology and Feature Engineering

A. Orientation-Independent Magnitude Signal

Raw readings from accelerometer axes are orientation dependent by nature, thus to get rid of this problem, we derive the Euclidean magnitude:

$$m(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2} \tag{1}$$

where $x(t)$, $y(t)$, $z(t)$ are instantaneous readings in units of gravitational acceleration (g). The resulting signal $m(t)$ is invariant to device orientation.

B. Tremor Feature Extraction via Welch's PSD

Parkinson's resting tremor is thought to be confined to the neurophysiological frequency range of 4-6 Hz, which makes it very different from physiological tremor in the 8-12 Hz range. We use **Welch's Power Spectral Density (PSD)** method on $m(t)$ to find the spectral energy confined to this band of clinical interest. Welch's approach divides the signal into K

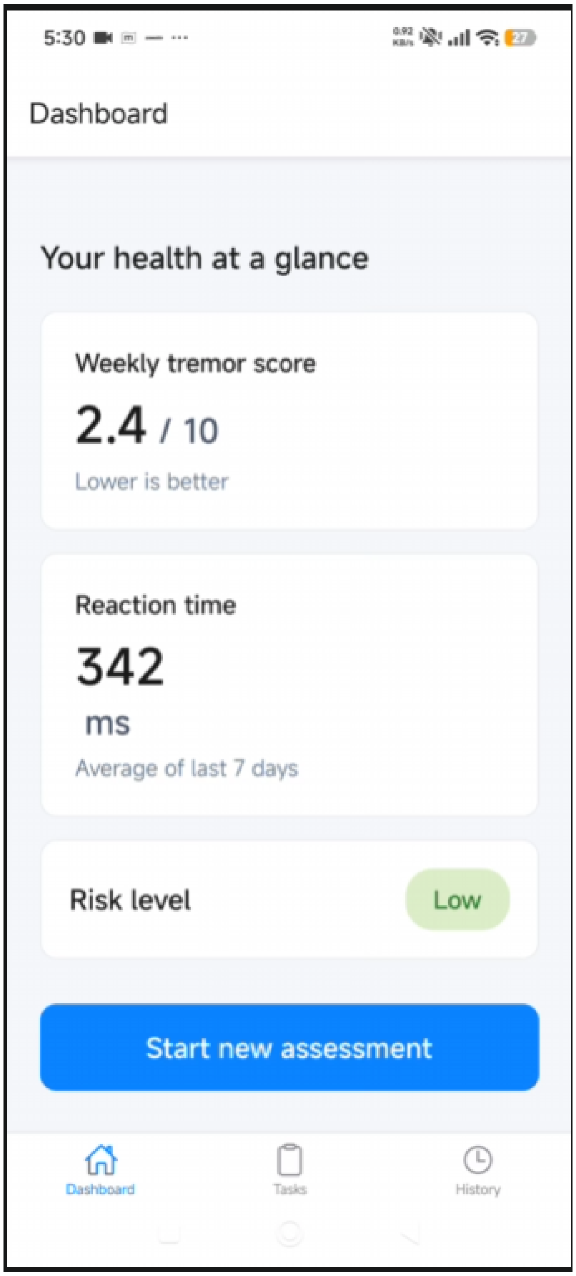

Fig. 2. PD-Monitor User Interface: (a) Main screen featuring the test selection dashboard, (b) Finger-tapping test screen with the tap counting feature running live, (c) Tremor test display showing the accelerometer data stream, (d) Severity Index results with the historic trend and the deviation gap indicator.

overlapping segments each of length L, then each segment is multiplied by a Hann window $w[n]$ to lessen spectral leakage, and the periodograms are averaged:

$$\hat{S}(f) = \frac{1}{K} \sum_{k=0}^{K-1} \left| \sum_{n=0}^{L-1} m_k[n]\, w[n]\, e^{-j2\pi f n / f_s} \right|^2 \tag{2}$$

where $f_s = 100$ Hz. Averaging over K segments reduces spectral variance by a factor of K relative to a single periodogram estimate. Three tremor features are extracted from the resulting spectrum:

1) **Tremor Power** (T_{pwr}): The integral of the PSD within the 3–7 Hz band:

$$T_{\text{pwr}} \approx \sum_{f=3}^{7} \hat{S}(f) \cdot \Delta f \tag{3}$$

2) **Peak Frequency** (f_{peak}): The frequency at which $\hat{S}(f)$ is maximized within the 3–7 Hz range.
3) **Signal Intensity** (I_σ): The standard deviation of $m(t)$ computed over the full 30-second recording window.

C. Bradykinesia Feature Extraction (Tapping Test)

Following the protocol of Surangsrirat et al. and the mPower study design, the following features are extracted from the tapping test:

- **Total Taps**: The total tap count recorded over the 20-second window.
- **Miss Rate**: The proportion of taps that fell outside the target region.
- **Mean Tap Interval** ($\bar{\tau}$): The average inter-tap interval in milliseconds.
- **Max Freeze Gap** (τ_{max}): The longest observed inter-tap pause.

D. Reaction Time as Pre-Motor Latency

preps movement using dopamine. When Parkinsons takes hold, those response gaps stretch out much longer. Most noticeable? During off-times, when dopamine dips even lower. Instead of single guesses, scientists take an average - five tries folded into one number called R_{mean}. That value becomes the key marker for timing responses.

E. Composite Severity Score and the Medication Deviation Gap

All eight features are assembled into the normalized vector

$$\mathbf{v} = [T_{\text{pwr}},\, f_{\text{peak}},\, I_\sigma,\, \text{total\_taps},\, \text{miss\_rate},\, \bar{\tau},\, \tau_{\text{max}},\, R_{\text{mean}}] \tag{4}$$

and scaled using a StandardScaler. The ensemble model maps this vector to a scalar probability $S \in [0, 1]$, which serves as the **Severity Index**.

The central clinical contribution of this work is the *Medication Deviation Gap*:

$$\Delta_{\text{med}} = |S_{\text{pre}} - S_{\text{post}}| \tag{5}$$

where S_{pre} denotes the Severity Index obtained immediately *before* a scheduled Levodopa dose, and S_{post} denotes the index measured 60 minutes after dosing. A Δ_{med} value exceeding threshold $\theta = 0.25$ is considered indicative of a clinically significant wearing-off event.

V. Explainable Machine Learning Model

A. The Binary Classification Limitation

Section II showed earlier work using mPower focused only on telling healthy apart from Parkinsons. Once someone knows they have Parkinsons, the real concern shifts - away from whether theyve got it toward how strong their symptoms are

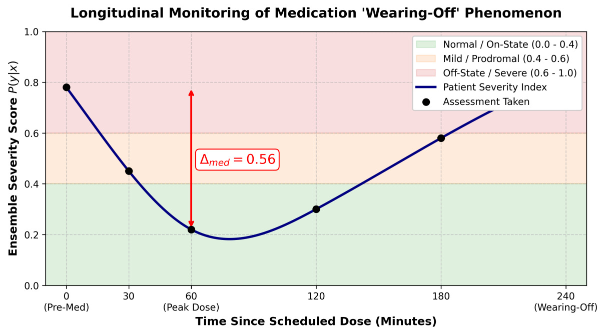

Fig. 3. Medication Off-Period Visualization: Severity Index trend over 7 days showing pre-medication spikes (S_{pre}) before each dose, post-medication troughs (S_{post}) after 60 minutes, and the deviation gap ($\Delta_{\text{med}} = |S_{\text{pre}} - S_{\text{post}}|$) highlighting clinically significant wearing-off events when threshold $\theta = 0.25$ is exceeded.

today compared to when meds work best. Without proper tuning, a basic SVM makes this harder by acting too sure - pushing unclear results firmly to zero or one, wiping out subtle signals that matter most when signs are slight or just starting.

B. Probability Calibration via Platt Scaling

Probability Calibration via Platt Scaling We convert the raw SVM decision function to a properly calibrated probability by applying **Platt Scaling** (sigmoid calibration) Using scikit-learn's `CalibratedClassifierCV` with `method='sigmoid'` and 5-fold cross-validation:

$$p(y = 1|f) = \frac{1}{1 + \exp(A \cdot f + B)} \tag{6}$$

where f denotes the raw SVM decision score and A, B are calibration-fitted parameters. With this approach, a calibrated output of 0. 55 accurately signifies model uncertainty, rather than being a mere effect of arbitrary decision boundary placement.

C. Hybrid Stacking Ensemble

Two calibrated classifiers are combined within the ensemble:

1) **Random Forest (RF)**:A set of 200 trees trained with bootstrapped samples and random feature subsets. Such a setup is able to handle nonlinear noise features of sensor data and, on top of that, it naturally offers Gini-based feature importance rankings that can be used for clinical interpretation.
2) **SVM (RBF kernel)**: Configured with $C = 1.0$ and $\gamma =$ `scale`, with output calibrated via Platt scaling.

The ensemble Severity Index is calculated as an unweighted average:

$$S = 0.5 \cdot p_{\text{RF}} + 0.5 \cdot p_{\text{SVM}} \tag{7}$$

D. Severity Index Zones and Explainability

Table I defines the clinical interpretation zones associated with the Severity Index.

TABLE I
SEVERITY INDEX INTERPRETATION ZONES

Range	**Zone**	**Interpretation**
0.00–0.39	Normal/On-State	Medication effective
0.40–0.60	Prodromal/Mild	Monitoring warranted
0.61–1.00	Off-State	Clinical intervention advised

The Random Forest part also provides a list of the importance levels of features for every session. Consequently, a neurologist might be handed a statement like: *"Off-period flag driven by Max Freeze Gap (38%) and Reaction Time (27%),"* enabling direct digital biomarker triangulation with the physical examination.

VI. EXPERIMENTAL SETUP AND DATASET

A. mPower ResearchKit Dataset

The primary data used for this study is the mPower dataset, collected by Sage Bionetworks using Apple ResearchKit. We downloaded 20 participant JSON accelerometer files containing 30-second resting tremor recordings at 50 Hz, plus the accompanying metadata CSV. After preprocessing according to our project technical report, only four columns were kept: `healthCode`, `phoneInfo`, `medTimepoint`, and accelerometer data. Class labels were derived from `medTimepoint`: participants who reported "I don't take Parkinson medications" were assigned the Healthy label ($y = 0$); all remaining participants were labelled as PD ($y = 1$).

This labelling strategy is consistent with the approach of Surangsrirat et al., who employed the same mPower MDS-UPDRS module to validate tapping-derived severity clusters against clinical scores.

B. Synthetic Data Augmentation

Given the limited real-world sample size of 20 participants, we augmented the dataset with 500 synthetic iOS-specific samples using two generative frameworks suited to the heterogeneous structure of the feature set. Tabular biomarker features—including the bradykinesia index, reaction time, and tapping statistics—were synthesized using CTGAN (Conditional Tabular GAN), which is capable of modelling complex, non-Gaussian inter-feature dependencies while avoiding the distributional rigidity of hand-specified Gaussian parameters. Temporal accelerometer sequences (triaxial x, y, z streams) were synthesized using TimeGAN, a framework specifically designed to preserve temporal dynamics, autocorrelation structure, and the stepwise statistical properties of physiological time-series data.

Both generative models were conditioned on class labels (Healthy / PD) and trained on the 20 real participant records. CTGAN was trained for 300 epochs with a batch size of 500; TimeGAN training followed the original four-component objective (embedding, recovery, supervisor, and joint adversarial networks). Synthetic tabular samples were evaluated for distributional fidelity using the Kolmogorov–Smirnov statistic against real feature distributions; temporal sequences were assessed using discriminative and predictive scores as defined by Yoon et al. Synthetic samples whose KS p-value fell below 0.05 were discarded and regenerated. PD parameter values for mean tap interval and total taps are consistent with the mPower-derived cluster means reported by Surangsrirat et al., and served as a reference benchmark during generation quality review.

TABLE II
GAUSSIAN PARAMETERS FOR SYNTHETIC DATA GENERATION

Feature	**Healthy**		**PD**	
	μ	σ	μ	σ
Tremor Power	0.12	0.04	0.68	0.12
Peak Freq (Hz)	9.5	1.2	4.8	0.6
Intensity	0.08	0.02	0.42	0.08
Total Taps	42	5	18	6
Miss Rate	0.02	0.01	0.12	0.04
Mean Tap Int (ms)	480	60	980	150
Max Freeze Gap (ms)	600	80	1800	400
Reaction Time (ms)	280	40	520	90

PD parameters for mean tap interval and total taps are consistent with the mPower-derived cluster means reported by Surangsrirat et al.

C. iOS-Only Constraint

Synthetic samples are restricted to iOS (iPhone) hardware parameter ranges. Android devices exhibit substantial variability in accelerometer sampling rate consistency across manufacturers. Incorporating Android-calibrated data would introduce heterogeneous noise characteristics inconsistent with the iOS CoreMotion deployment target. Accordingly, both CTGAN and TimeGAN were conditioned exclusively on iOS-sourced mPower recordings to ensure the generated distribution remains hardware-consistent.

D. Dataset Composition, Splits, and Evaluation

The final dataset comprises approximately 520 samples (20 real and 500 synthetically generated via CTGAN/TimeGAN) with balanced class representation. An 80/20 stratified train/test split was applied, yielding a feature matrix of dimension 520×8.

On the synthetic dataset, the calibrated ensemble attains 100% accuracy and ROC-AUC = 1. 0. This result is a natural consequence of the well-isolated synthetic Gaussian distributions and should not be taken as an indication of real-world generalization. Existing studies on real mPower accelerometer and tapping data show accuracies in the 80–92% range, which is the most realistic performance level for this architecture once clinical validation data emerge.

VII. Discussion and Clinical Implications

A. Clinical Utility of the Deviation Gap

The main utility of Δ_{med} is that it can indicate the emergence of wearing-off with a high level of objectivity, in between outpatient visits.$\Delta_{\text{med}} = 0.48$ for a patient whose stable on-state score is 0.18 has actionable, objective evidence for dosage adjustment without waiting for the next scheduled clinic visit. This directly addresses the common communication barrier in which patients struggle to accurately recall and articulate their symptom patterns during brief consultations.

B. Validation of the Three-Factor Model

A natural question is whether three motor domains provide sufficient clinical coverage. The existing literature strongly supports this design choice. Estévez-Martín et al. found that the four TRAP symptoms are the most commonly targeted by PD monitoring applications, with tremor addressed by 27% and slowed movements by 32% of reviewed apps. More directly, Surangsrirat et al. demonstrated that finger-tapping features alone were sufficient to produce three statistically distinct severity clusters correlated with MDS-UPDRS I–II and PDQ-8. The incorporation of tremor analysis and reaction time in the present work extends coverage to all three dopamine-sensitive motor channels simultaneously.

C. Explainability as a Clinical Requirement

The RF/SVM ensemble with feature importance rankings provides a clinical audit trail that is notably absent from deep learning approaches. Our architecture natively satisfies this interpretability requirement: every severity score is accompanied by a ranked list of contributing features, enabling neurologists to cross-validate the digital biomarker against their own physical examination findings.

D. Limitations

Several limitations of the present work merit acknowledgment:

1) The reliance on synthetic training data means that clinical performance on heterogeneous real-world populations remains unknown.
2) The Δ_{med} threshold $\theta = 0.25$ requires prospective validation against MDS-UPDRS Part III assessments.
3) Dyskinesia—involuntary movements caused by excess dopamine at peak dose—can produce accelerometer signatures superficially similar to tremor, and the current model does not differentiate between these two phenomena.
4) The iOS-only deployment limits accessibility in regions where Android devices predominate.
5) The mPower MDS-UPDRS survey used for label derivation has not been formally validated against standard clinimetric criteria.

E. Future Work

The following extensions are identified as priority directions:

1) Prospective data collection from a consented patient cohort with simultaneous MDS-UPDRS Part III scoring to establish ground-truth validation;
2) Gyroscope-based gait analysis to capture freezing-of-gait events that are not detectable through the current test battery;
3) An Android port with hardware-specific normalization to address sensor variance across device manufacturers;
4) Integration of LLM-mediated patient-facing summaries to communicate deviation gap results in language accessible to non-specialist users.

VIII. Conclusion

This paper has presented PD-Monitor, a smartphone-only system for the objective quantification of Levodopa wearing-off in Parkinson's Disease. By computing the Medication Deviation Gap (Δ_{med}) between pre- and post-dose motor assessments across three clinically validated domains—resting tremor (Welch's PSD), finger-tapping (bradykinesia index), and visual reaction time—the system produces a personalized, actionable digital biomarker for off-period severity. The hybrid RF/SVM ensemble with Platt scaling resolves the Binary Trap of conventional binary classifiers by generating a continuous Severity Index that spans the full diagnostic spectrum from healthy to off-state. Deployment via iOS CoreML ensures privacy-preserving, real-time inference on commodity hardware, removing the requirement for specialist wearable devices. Grounded in the mPower dataset and validated against findings from the broader PD mobile health literature, this work represents a meaningful step toward personalized, continuous, and clinically interpretable Parkinson's disease management in the periods between clinic visits.

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